

Cross-Chain Content Rights Management Service: A Unified Framework for Decentralized Subscription, Pay-Per-View, and Purchase Models

Introduction

The digital content industry has emerged as one of the most rapidly expanding sectors within the global economy. In 2024, worldwide revenues from digital media and entertainment surpassed USD 550 billion, with forecasts predicting an increase to approximately USD 750 billion by 2028 (PwC Global Entertainment & Media Outlook 2024–2028). Streaming platforms, digital publishing, online education, independent journalism, and marketplaces for user-generated content now serve as the primary channels whereby creators engage with audiences. Nevertheless, despite this remarkable expansion, creators continue to receive only a modest portion of the value they generate. On the predominant centralized platforms—such as YouTube, Spotify, Netflix, Substack, Patreon, and Apple—intermediaries routinely retain between 30% and 45% of gross revenue through platform fees, payment processor charges, and revenue-sharing agreements. Furthermore, creators face geographical restrictions, delayed payouts (often 60-90 days), arbitrary content demonetization, and a complete lack of portability for their audience relationships.

The emergence of blockchain technology and non-fungible tokens (NFTs) in 2021–2022 was widely anticipated to revolutionize this model by facilitating direct, verifiable digital ownership and programmable royalties. While NFTs effectively demonstrated that scarcity and provenance can be enforced on-chain, the vast majority of NFT initiatives from 2021 to 2025 remained speculative collectibles rather than practical tools for content distribution. A comprehensive bibliometric study published in May 2025, analyzing 2,847 academic papers concerning blockchain and music, revealed that 79.4% concentrated solely on copyright registration, watermarking, or anti-piracy tracing, while only 28.2% examined actual revenue-distribution mechanisms (SET Journal, 2025). Even among utility-centric projects such as Audius, Mirror.xyz, Sound Protocol, Livepeer, and various NFT marketplaces, no existing production system provides creators with a rights primitive capable of simultaneously addressing three principal consumer preferences: recurring subscription access, pay-per-view or per-segment micropayments, and outright permanent ownership.

This fragmentation compels creators to make challenging trade-offs. A musician seeking to offer monthly subscriptions via Mirror, sell individual tracks as NFTs on Sound Protocol, and facilitate pay-per-view concert streams on Theta must manage three separate smart contract suites, three distinct user onboarding processes, and three divergent economic models—each characterized by its unique fee structure, technical considerations, and community segmentation. The issue becomes more pronounced when creators operate across multiple blockchains. Although cross-chain interoperability has advanced considerably with the 2025 releases of IEEE Std 3221.01, Chainlink CCIP v2, IBC v2 (now connecting over 115 sovereign chains), and Polkadot's XCM v5, these standards were primarily conceived for fungible assets and basic NFTs. Metadata pertaining to rich content—such as royalty distribution vectors, geographic licensing regulations, collaborator lists, and consumption counters—frequently surpass practical payload limits of approximately 4 KB imposed by XCM and IBC. This often compels developers to fragment policy data or rely on centralized oracles, thereby reintroducing trust assumptions that blockchain technology was originally designed to eliminate.

The economic ramifications are considerable. Bridge-based cross-chain solutions in 2025 still incur cumulative fees of 8–15%, and exhibit failure rates of 12–18% during periods of peak congestion (Chainalysis Bridge Report Q3 2025). Micro-transaction costs on Ethereum Layer-2 networks range from USD 0.01 to USD 0.08, rendering genuine per-minute or per-segment video monetization economically unviable for independent creators. Meanwhile, centralized incumbents continue to dominate subscription revenue, accounting for 68% of all streaming income, according to the RIAA and IFPI 2025 reports, because they can deliver sub-second access decisions and a consistent user experience across devices.

This thesis directly addresses the identified gap by proposing a shared-security, multi-chain content rights management framework built on Polkadot’s relay-chain architecture, XCM cross-consensus messaging, and ink! smart contracts written in Rust. *The core innovation is a unified rights token.* This single on-chain data structure can simultaneously represent an active subscription with auto-renewal logic, a pay-per-view consumption counter, and permanent ownership with programmable royalty distribution. By leveraging Polkadot’s pooled validator security, native message-passing capabilities, and the memory-safety guarantees of Rust, the framework aims for sub-second finality in access verification, transaction costs below one cent for micro-payments, and creator revenue retention exceeding 95%—performance thresholds that are unattainable by any existing centralized or decentralized system across all three monetization models concurrently.

The research is driven by four converging macro-trends observed throughout 2025:

- The rapid emergence of independent creator economies and the increasing demand for equitable revenue sharing.
- The development of heterogeneous shared-security blockchains, including Polkadot, Cosmos, and their respective ecosystems.
- Increasing regulatory scrutiny of tokenized assets under the upcoming EU MiCA (2026) and U.S. SEC frameworks.
- Widespread consumer fatigue arising from fragmented subscription services across walled gardens.

The contribution is therefore both theoretical, providing a new reference framework for second-generation Web3 content monetization, and practical, including an open-source prototype that music labels, streaming services, educational platforms, journalists, and individual creators can deploy or fork without sacrificing user experience or decentralization.

Background and Related Work

1 Evolution of Digital Rights Management

Traditional DRM systems, such as Adobe Content Server, Google Widevine, Apple FairPlay, and Microsoft PlayReady, rely on centralized license servers and cryptographic envelopes to control access. Although these systems are effective on a broad scale, they are frequently characterized by opacity, vendor lock-in, geographic restrictions, and significant revenue extraction, which ranges from 20% to 45%. In response, several blockchain-based alternatives have been developed through three distinct innovations.

- **First wave (2016–2019):** Bitcoin-colored coins and Ethereum ERC-721 prototypes focused on provenance and anti-piracy.
- **Second wave (2020–2023):** Academic literature concentrated on copyright registration and watermarking (Xie & Liu 2022; Garba et al. 2020; Zhang et al. 2024).
- **Third wave (2024–2025):** Industry deployments shifted toward programmable royalties and fractional ownership (Eluvio Content Fabric, Audius, Livepeer, Unique Network).

Despite progress, a 2025 bibliometric review of 2,847 blockchain-music papers found that 79.4% address only protection mechanisms, while just 28.2% explore revenue distribution (SET Journal, May 2025).

2 Cross-Chain Interoperability Standards (2025)

Four interoperability paradigms dominate the current landscape:

1. Relay-chain/shared-security models (Polkadot, Cosmos Hub)
2. Light-client bridges (Snowbridge, Hyperbridge, Rainbow Bridge)
3. Oracle networks (Chainlink CCIP v2)
4. Hash-time-locked contracts and atomic swaps

The IEEE published Standard 3221.01-2025, the inaugural formal standard for cross-chain transaction consistency. However, its scope explicitly excludes complex metadata payloads necessary for content licensing. XCM v5 (Polkadot), IBC v2 (Cosmos, encompassing over 115 chains), and CCIP v2 all exhibit practical message size limitations of 2 to 4 KB, which are inadequate for detailed rights expressions (Unique Network 2025; Calibraint 2025).

3 Existing Decentralized Monetization Attempts

- NFT marketplaces such as OpenSea, Rarible, and Unique Network excel in providing permanent ownership and secondary royalties; however, they do not offer native subscription primitives.
- Streaming protocols such as Audius, Theta, and Livepeer incorporate token incentives; however, they remain limited to individual ecosystems.
- Experiments centered on subscriptions (Mirror.xyz, Friends With Benefits) depend on the periodic issuance of access tokens—an expensive approach that incurs costs ranging from USD 0.05 to 0.20 per renewal on Ethereum Layer 2 solutions.
- Hybrid platforms (Eluvio Content Fabric 2024; Fourchain 2025) integrate centralized delivery mechanisms with blockchain settlement processes, thereby sacrificing complete decentralization.

No reviewed system (112 sources analysed) implements a single rights token capable of natively enforcing subscription auto-renewal, pay-per-view counters, and permanent ownership with cross-chain royalty propagation.

4 Polkadot Ecosystem as Enabling Substrate

Polkadot's shared-security model eliminates the need for individual parachains to bootstrap their validator sets, thereby significantly reducing the attack surface compared to sovereign layer 2 solutions (Crippa & Polu 2022). The launch of Agile Coretime in 2025 introduced market-based resource allocation, while XCM v5 incorporated scheduled locations and barrier instructions—features that explicitly facilitate recurring cross-chain payments. The implementation of ink! v6 with PolkaVM offers execution speeds that are 5 to 10 times faster than Solidity for identical logic, and Rust's inherent memory safety features eliminate entire categories of reentrancy and overflow vulnerabilities, which have previously cost the Ethereum ecosystem billions (Parity Technologies 2025).

Research Questions and Objectives

5 Primary Research Question (Refined November 2025)

How can a shared-security, multi-chain framework built on Polkadot's XCM and ink! smart contracts deliver a unified rights token that natively supports recurring subscriptions, pay-per-view micro-transactions, and permanent ownership transfers across heterogeneous blockchain networks, while achieving sub-second finality, sub-cent transaction costs, and creator revenue retention greater than 95%?

6 Sub-Questions and Objectives (Refined November 2025)

The refined sub-questions and corresponding objectives are presented below:

SQ1 / RO1 – Cross-chain recurring payments and metadata

How can XCM v5+ be extended to support scheduled subscription renewals and rich rights metadata with >99% atomic success?

→ Design and prototype XCM extensions achieving <2s finality.

SQ2 / RO2 – User experience versus decentralization balance

Which ink-based design patterns facilitate access verification within one second while ensuring HHI remains below 1,500?

→ Implement a unified rights token with chain extensions and off-chain indexing.

SQ3 / RO3–6 – Hybrid monetization model

Is it feasible for a single on-chain rights object to enforce all three models with automated cross-chain royalties, and what are the potential economic benefits?

→ Consolidate subscription, pay-per-view (PPV), and purchase mechanisms into a single composable module; establish benchmarking against 2025 baseline standards; integrate optional

zero-knowledge (ZK) compliance interfaces; and quantify the anticipated savings through Monte Carlo simulation.

These refined questions and objectives directly address the critical gaps identified in the literature (G1–G7) and provide clear, measurable success criteria for the remainder of the research.

Top 25 Annotated Bibliography

1. **IEEE** (2025). *IEEE Std 3221.01-2025: Standard for blockchain interoperability – Cross-chain transaction consistency protocol*. IEEE Standards Association.

The first formal international standard for cross-chain atomicity and consistency. Serves as the primary evaluation benchmark for success rate and latency of rights transfers.
2. **Unique Network** (2025). *Composable Subscriptions & NFTs using XCM* (blog series + technical papers, Jan–Sep 2025). <https://unique.network/blog>

Closest prior art in the Polkadot ecosystem; demonstrates recurring cross-chain NFTs. Lacks PPV counters and unified hybrid model → the thesis directly extends this work.
3. **MarketsandMarkets** (2025). *Blockchain Digital Rights Management Market – Global Forecast to 2029*.

Projects market size from USD 0.25 B (2025) to USD 1.42 B (2029) at 54.2% CAGR. Primary economic justification for the research.
4. **Zhang, L., Wang, H., & Chen, M.** (2024). A digital resource copyright protection scheme based on blockchain cross-chain technology. *Heliyon*, 10(16), e34781.

State-of-the-art academic cross-chain DRM (2024). Focuses on registration and enforcement but not monetization → gap G1.
5. **USPTO & USCO** (2024). *Non-Fungible Tokens and Intellectual Property: Final Report*. Joint study.

Authoritative legal analysis of NFT ownership, licensing, and royalties. Legal foundation for the PurchaseVerification logic.
6. **Wood, G.** (2022). *XCM: The cross-consensus message format*. Polkadot Blog (updated 2025).

Core specification of the messaging layer used for all cross-chain rights transfers.
7. **Parity Technologies** (2025). *ink! v6 Documentation & PolkaVM Release Notes*.

Primary smart-contract language and runtime. Rust safety + 5–10× performance over Solidity.
8. **Chainlink** (2025). *Cross-Chain Interoperability Protocol (CCIP) Technical Whitepaper v2*.

Leading oracle-based bridge solution; used as comparison for external-chain integration.
9. **IBC Protocol Team** (2025). *IBC v2 Specification and Launch Report*. <https://ibcprotocol.dev>

Sovereign interoperability standard connecting 115+ chains (March 2025 launch).

10. **Xie, R., & Liu, Y.** (2022). Research on copyright protection of digital works based on multi-blockchain. *Recent Advances in Computer Science and Communications*, 15(8).
Early multi-blockchain copyright framework; reduced block times vs single chain.
11. **Garba, A. et al.** (2020). A digital rights management system based on a scalable blockchain. *Peer-to-Peer Networking and Applications*, 14(5).
Scalable watermarking + transaction model; single-chain limitation.
12. **Eluvio** (2024). *The Content Fabric: Multi-Chain Content Ownership and Distribution Whitepaper*.
Industry competitor with marketplace-ready hybrid fabric.
13. **Calibraint** (2025). *Advanced Strategies for NFT Marketplace Architecture*.
Cross-chain marketplace design claiming 70% fee reduction via bridges.
14. **Fourchain & Oodles Blockchain** (2025). *Cross-Chain NFT Marketplace Development Reports*.
Real-world 2025 deployments supporting 5–10 chains; purchase-only focus.
15. **Frontiers in Blockchain** (2025). Regulatory barriers in blockchain-based IP trade: Evidence from China.
Identifies regulatory compliance as the single largest barrier (weight 0.308).
16. **SET Journal** (2025). Bibliometric analysis of blockchain and music (2,847 papers).
79.4% protection-only, 28.2% revenue distribution → quantitative proof of gap G1.
17. **PwC** (2024). *Global Entertainment & Media Outlook 2024–2028*.
Primary source for digital content market size and subscription dominance (68%).
18. **Chainalysis** (2025). *Bridge Exploit Report Q3 2025*.
Documents >\$2.1 B lost in bridge hacks 2024–2025; justification for trust-minimized design.
19. **Crippa, L., & Polu, S.** (2022). Analysis of Polkadot: Architecture, internals, and contradictions. *IEEE Blockchain 2022*.
Critical technical analysis of shared-security trade-offs.
20. **Habermeier, R. et al.** (2021). *Polkadot's messaging scheme*. Web3 Foundation.
Original XCM paper; foundational for scheduled locations extension.
21. **Parity Technologies** (2025). *Agile Coretime Whitepaper*.
Market-based parachain resource allocation launched 2025.
22. **Snowbridge & Hyperbridge Teams** (2025). *Trust-minimized Ethereum–Polkadot Bridge Specifications*.
Production bridges used for external-chain integration.

23. **RIAA & IFPI** (2025). *Global Music Report 2025*.

Subscription revenue share and creator payout statistics.

24. **USC Annenberg** (2024). *Creator Economy Report 2024*.

Platform fee ranges (30–45%) and delayed payouts.

25. **Di Francesco, A., & Zoppi, S.** (2024). A survey on trustless cross-chain interoperability solutions. *DLT 2024 Proceedings*.

Comprehensive taxonomy including Polkadot strengths and remaining gaps.

Implementation Notes (Phase 4–5)

This thesis draft was written in November 2025 during Phase 1, before implementation. The following claims require revision in the final thesis to reflect measured results from Phase 4–5 testing.

Performance Claims Requiring Revision

Table 1: Draft Claims vs Measured Results

Draft Claim	Measured Result	Revision Needed
“sub-second finality in access verification”	~6 seconds (one block time)	Replace with “deterministic ~6 second inclusion latency”
“XCM extensions achieving <2s finality” (RO1)	18–32 seconds (3–5 parachain blocks)	Replace with measured XCM latency range
“access verification within one second” (RO2)	~6 seconds	Replace with measured block time; note client-side caching as path to sub-second
“transaction costs below one cent”	Achieved—on-chain fees are negligible	No revision needed
“creator revenue retention exceeding 95%”	100% in self-publishing model (configurable royalty splits)	Update to reflect self-publishing eliminates intermediary entirely

Architecture Claims Requiring Revision

Table 2: Architecture: Draft vs Actual

Draft Claim	Actual Implementation	Revision Needed
“ink! smart contracts deliver a unified rights token”	<code>pallet-content-rights</code> (FRAME pallet) is the primary logic; ink! provides API layer via <code>pallet-revive</code> precompile	Reframe: FRAME pallet as primary, ink! as external interface
“XCM cross-consensus messaging” for rights transfers	Confirmed—6 XCM extrinsics, 100% success rate	No revision needed
“zero-knowledge (ZK) compliance interfaces”	Not implemented	Move to Future Work
“quantify savings through Monte Carlo simulation”	Not implemented; replaced by direct centralized benchmark ($270\times$ TPS ratio)	Replace Monte Carlo with comparative analysis methodology
“HHI remains below 1,500”	Not measured	Remove or move to Future Work

What the Thesis Should Emphasise

The final thesis should reframe the contribution around what was actually demonstrated:

1. **Unified rights pallet** with subscription, PPV, and ownership in a single FRAME pallet (17 extrinsics, 56 unit tests)
2. **Cross-chain operations** via XCM with paid execution (100% success, 18–32s latency)
3. **Ethereum bridge** via full Snowbridge v1 E2E (2 ETH bridged with cryptographic proof verification)
4. **Automatic royalty propagation** with configurable splits (up to 10 collaborators, basis-point precision)
5. **On-chain auto-renewal** via `on_initialize` scheduler pattern
6. **Performance trade-off analysis**: $270\times$ slower than centralized but trustless, censorship-resistant, and self-publishing
7. **Self-publishing model** that eliminates intermediaries entirely (vs traditional 15–30% platform fees)